

Can weaker coupling produce intermittent gamma synchronization?

exp086 · 19 August 2026 · Draft

Abstract

In macaque V1, nearby gamma rhythms did not hold one fixed relative-phase position. Their relative phase continued to move, but its velocity slowed near positions where coupling made the two instantaneous frequencies more similar. This phase-dependent slowing concentrated the observed phase differences around those positions, producing a peaked distribution despite continuing phase slips [1]. The effect could create recurring windows for effective communication between cortical populations without forcing them into permanent synchronization. Starting from two rigidly phase-locked PING networks, this experiment reduces coupling and asks whether their relative phase begins to circulate while still slowing near a preferred position. Network structure, input, frequency difference, delay, and noise remain fixed. Broader parameter maps and oscillator modelling are left for later experiments.

Prior work

Lowet et al. (2015) mapped synchronization across detuning and coupling in two 80-E, 20-I PING networks with reciprocal E-to-E and E-to-I fan-in eight [2]. Lowet et al. (2017) found the corresponding signature in macaque V1: coupling reduced instantaneous frequency differences near preferred phases, producing imperfect rather than permanent synchronization [1].

The shared parameter-space axes are uncoupled frequency detuning Δf and measured effective interaction strength ε . Mean frequency $|f|$ sets the timescale; phase noise controls how sharply the rhythms lock.

Quantity	Lowet regime	Our starting point	First move
Mean frequency $ f $	Approximately 30–40 Hz in the illustrated model and macaque conditions	48.7 Hz	Hold fixed initially
Detuning Δf	Illustrated macaque examples span approximately 2.8–4.8 Hz	4 Hz, or 0.082 of $ f $	Hold near the starting value

Effective interaction ε	Phase-dependent frequency modulation is approximately 1.6–1.8 Hz in representative macaque conditions	Not measured; the interaction cancels 4 Hz at the locked phase	Describe the correction; fit ε later
Synchronization	Representative phase-locking values approximately 0.3–0.5 with phase slips	Phase concentration 0.997 with no late slips	Weaken coupling first

Coupling units. Both models divide total pathway strength across eight afferents. Lowet reports summed conductance density in mS/cm²; SNNLANG reports summed absolute conductance in μ S. Normalizing each by target leak conductance places our E-to-E and E-to-I strengths at $\kappa = 1.6$ and 0.8, within Lowet’s broad sweep but more E-to-E dominated [2]. Here κ is cross-network conductance divided by target leak conductance.

Methods

This experiment has not been run. Its starting values and topology were sourced from exp085 and are fixed here as design inputs.

1. **Define the starting rhythms.** Each network contains an excitatory population, an inhibitory population, and a local E-to-I-to-E feedback loop. Reciprocal excitation targets both E and I populations with $K_{EE} = 0.08 \mu$ S, $K_{EI} = 0.08 \mu$ S, and delay $d = 2$ ms. Drive Networks A and B at 300 and 260 Hz. Run them without cross-network coupling and confirm regular rhythms near mean frequency 48.7 Hz and detuning 4 Hz. Interpolate phase between consecutive excitatory volleys and calculate their wrapped and unwrapped relative phase. Generates Result 1.

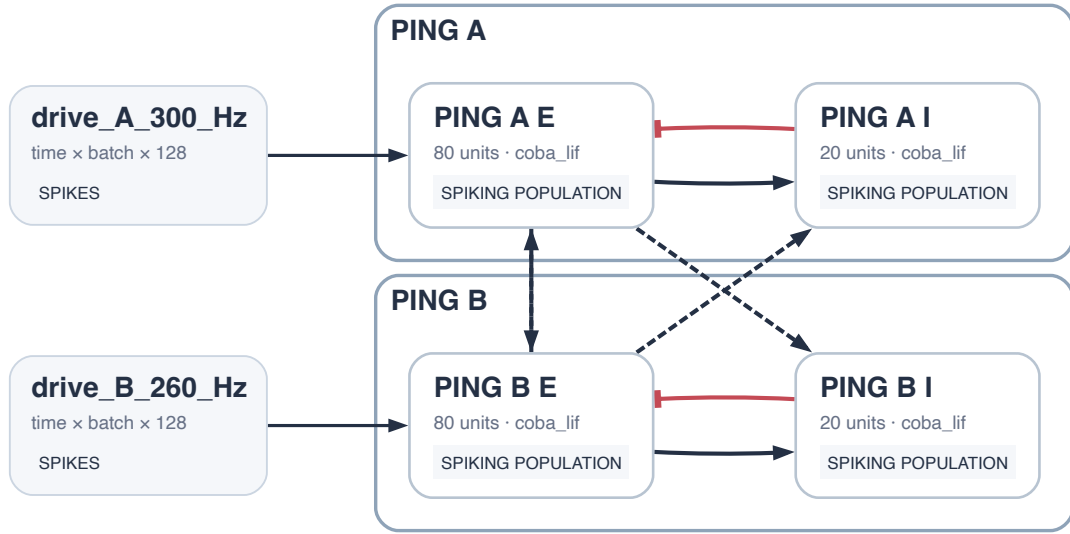


Figure 1: Planned starting topology. Networks A and B each contain a local E-to-I-to-E PING loop. Reciprocal projections target E and I populations with independently controlled strengths.

2. **Reduce coupling while everything else stays the same.** Save the network immediately before coupling begins. Replay from that point several times with the same neuron states and the same pre-generated input spike trains after that point. Change only the coupling strength K : begin at $0.08 \mu\text{S}$, then use progressively weaker values down to zero. E-to-E and E-to-I coupling always use the same value. Run one trajectory at each value of K ; repeated seeds and trials are outside this experiment. For each coupling strength, track the phase gap between the two rhythms. Strong coupling may keep that gap fixed. Weaker coupling may allow one rhythm to repeatedly gain a full cycle on the other, producing phase slips. We are looking for the intermediate behaviour reported by Lowet et al. (2017): phase slips continue, but the phase gap repeatedly slows near one preferred value, making that value more common [1]. Select one trajectory that shows this pattern. This demonstrates the behaviour once; it does not establish reliability across seeds. Generates Result 2.

What weakening coupling should change

Design schematic — these curves are not simulated results

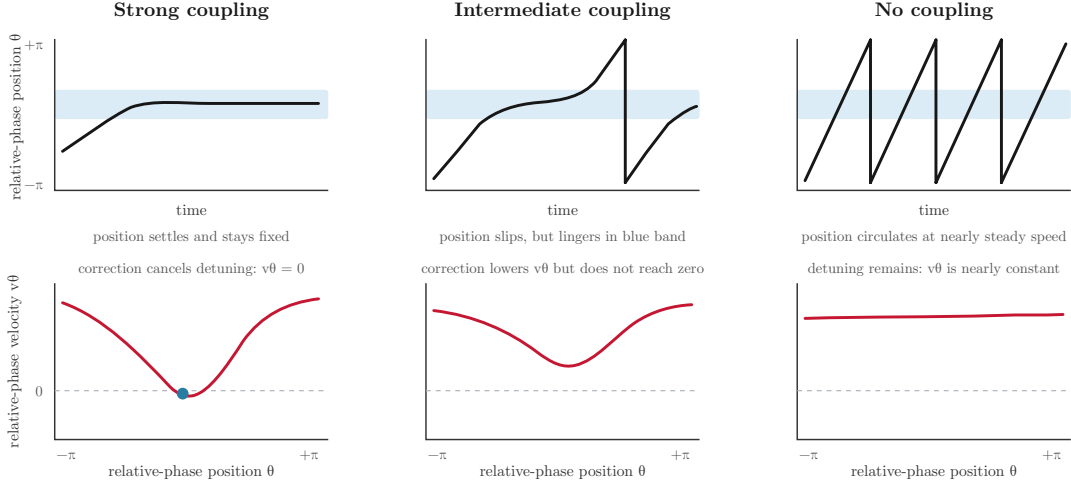


Figure 2: **Expected mechanism and effect of weakening coupling.** Strong coupling can fully cancel the networks' frequency difference and hold a fixed relative-phase position. At intermediate coupling, the correction is too weak to stop relative-phase motion but still slows it near a preferred position. With no coupling, the original frequency difference produces nearly uniform relative-phase drift. These curves illustrate the experimental logic; they are not simulated data. Result 2 will reproduce this layout with measured trajectories.

3. **Relate phase velocity to phase position.** For the selected condition, estimate each network's instantaneous frequency from consecutive excitatory volleys. Relative-phase velocity is $v_\theta = 2\pi(f_A - f_B)$, where f_A and f_B are the instantaneous frequencies of Networks A and B. Group v_θ by wrapped relative-phase position θ , and compare the resulting velocity curve with the distribution of θ . Intermittent attraction requires continued phase slips, a non-uniform phase distribution, and lower absolute velocity near the distribution's preferred position. Generates Result 3.

Results

1. Uncoupled rhythms.

- **Axes.** Time on the horizontal axis; normalized population activity on the vertical axis.
- **Traces.** Excitatory and inhibitory activity from Networks A and B, followed by their wrapped relative-phase position.
- **Why.** Confirm that both networks generate regular gamma rhythms with a controlled intrinsic frequency difference before coupling.
- **Expectation.** Each network maintains a clean PING rhythm while their uncoupled relative phase repeatedly wraps.

2. Coupling boundary.

- **Axes.** Reproduce the three-column layout of the Method 2 schematic using measured data. The upper row shows time against wrapped relative-phase position. The lower row shows relative-phase position against mean relative-phase velocity.

- **Traces.** Show the strong starting coupling, the selected intermediate coupling, and zero coupling. State the measured value of K above each column.
- **Why.** Show the observed transition in the same visual language used to explain the proposed mechanism.
- **Expectation.** Strong coupling holds a fixed relative-phase position. Intermediate coupling permits slips but slows near a preferred position. With no coupling, relative phase drifts at nearly constant velocity.

3. Intermittent phase attraction.

- **Axes.** The first two panels show time against relative-phase position θ and velocity v_θ . The third shows position θ against mean velocity v_θ . The fourth shows the distribution of θ .
- **Traces.** Wrapped phase position and phase velocity through time, followed by the position-binned velocity curve and phase-position distribution for the same condition.
- **Why.** Test the specific Lowet 2017 signature without fitting a full oscillator model or mapping a second parameter.
- **Expectation.** Position continues to wrap through phase slips, but absolute velocity is smallest near the position at which the phase distribution peaks.

References

References

1. Lowet, E., Roberts, M. J., Peter, A., Gips, B., and De Weerd, P. (2017). A quantitative theory of gamma synchronization in macaque V1. doi:10.7554/eLife.26642
2. Lowet, E., Roberts, M., Hadjipapas, A., Peter, A., van der Eerden, J., and De Weerd, P. (2015). Input-dependent frequency modulation of cortical gamma oscillations shapes spatial synchronization and enables phase coding. doi:10.1371/journal.pcbi.1004072